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Quiz: Wireless Transmission - Channel Capacity (Signal vs. Interference)

Started at	Sunday, April 10, 2022, 4:10 p.m
status	Completed
Finished at	Sunday, April 10, 2022 at 11:59 p.m
Spent time	7 hours 48 minutes
valuation	**8.33 out** of 10.00 (**83** %)
feedback	Congratulations, your performance was very good (80%-90%).

question **1**

Complete

Score 0.33 out of 1.00

From what aspects does Shannon abstract from?

Choose one or more answers:

- [x] a. Receiver induced noise
- [ ] b. receiver sensitivity
- [ ] c. coding scheme
- [ ] i.e. Number of antennas
- [ ] e. modulation scheme

question **2**

Complete

Score 1.00 out of 1.00

How can Shannon's law be use to estimate the capacity in interference-limited system?

Choose one or more answers:

- [x] a. We assume interference to be similar to noise and apply the same formula replacing noise by interference
- [x] b. We assume that interference is much higher than noise so that noise can be neglected

question 3

Complete

Score 1.00 out of 1.00

How can the capacity in interference-limited systems be increased further?

Choose one or more answers:

- [x] a. Add more frequency bands
- [ ] b. Increase the transmit power
- [x] c. Add more sectors to the given base station infrastructure to decrease pathloss while decreasing interference
- [x] i.e. Add more base stations to reduce the pathloss
- [ ] e. Decrease the transmit power

Frage 4

Vollständig

Erreichte Punkte 1,00 von 1,00

Impact of channel bandwidth, received signal strength and noise on the capacity of a channel: are there approximations for these factors?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. No
- [x] b. Yes, Shannon´s law
- [ ] c. Yes, Additive White Gaussian Noise (AWGN)

Frage **5**

Vollständig

Erreichte Punkte 1,00 von 1,00

The following factors affect the quality of reception:

Wählen Sie eine oder mehrere Antworten:

- [x] a. Coding scheme
- [x] b. Signal bandwidth
- [x] c. Receiver-induced noise
- [x] d. Pathloss of radio channel from sender to receiver
- [x] e. Multipath propagation

Frage **6**

Vollständig

Erreichte Punkte 1,00 von 1,00

What are core results of Shannon´s capacity law?

Wählen Sie eine oder mehrere Antworten:

- [x] a. Signal and Noise are reciprocal of each other
- [x] b. The channel capacity is proportional to the channel bandwidth

Frage **7**

Vollständig

Erreichte Punkte 1,00 von 1,00

What defines interference-limited systems?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. The increase in transmit power of all transmitters does result in higher capacity overall
- [x] b. The increase of the transmit power of a single terminal does increases its capacity
- [x] c. The addition of new transmitters to the system introduces so much interference to others, that there is no benefit for the overall system capacity

Frage **8**

Vollständig

Erreichte Punkte 1,00 von 1,00

What factors influence the quality of the reception at a given receiver's input?

Wählen Sie eine oder mehrere Antworten:

- [x] a. Intersymbol interference
- [x] b. Modulation scheme
- [x] c. Interference from others
- [x] d. Strength of the signal as such
- [x] e. Noise

Frage **9**

Vollständig

Erreichte Punkte 1,00 von 1,00

What further ideas can be employed to increase the capacity in interference-limited systems?

Wählen Sie eine oder mehrere Antworten:

- [x] a. Use interference-cancelation techniques
- [x] b. Use directional antennas at the terminals
- [x] c. Use D2D communication where applicable, i.e. between terminals that are close together

Frage **10**

Vollständig

Erreichte Punkte 0,00 von 1,00

What is not covered by Shannon´s capacity law?

Wählen Sie eine oder mehrere Antworten:

- [ ] a. Intersymbol interference
- [x] b. Interference from third parties
- [x] c. Noise other than additive white Gaussian noise

Vorhergehende Aktivität

◀ Quiz: Wireless Transmission - Antennas

Direkt zu:

next activity

Quiz: Wireless Transmission - Multiplexing (Splitting up the Radio Resource) ▶